

Exploring LLM Agents as Seamless and Context-aware UI Adjusters for the Visually Impaired

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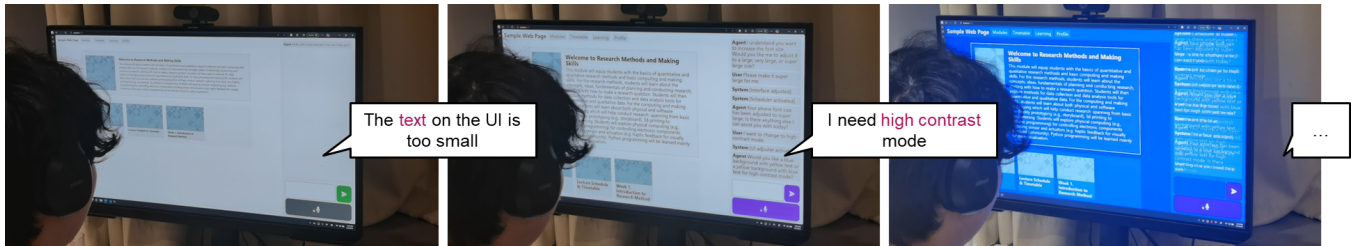


Figure 1: LLM-powered seamless and context-aware UI adjustments: the VI user can naturally speak to the conversational agent for various interface accommodations, the agent is capable of understanding the user's conditions, inferring their needs, and taking actions to achieve adaptive UI. Based on the device and ambient context, such as elements on the UI, screen size, and ambient light, the agent will adjust the UI differently.

Abstract

The user interface (UI) of traditional electronic devices often lack intuitive and flexible methods to accommodate the diverse needs of visually impaired (VI) users. This paper explores the role of Large Language Model (LLM)-powered Conversational Agents (CAs) serving as intelligent, context-aware mediators to achieve VI-friendly adaptive UI. Our approach focuses on facilitating real-time UI adjustments based on human-AI conversations and device and ambient context, leveraging LLM CAs' reasoning and tool-using capabilities to accommodate different types of visual impairments. A prototype was developed to demonstrate how the system can dynamically adapt the interface. Our initial observations suggest that this approach has the potential to promote a seamless and inclusive experience for VI users, particularly on daily information terminals that are not flexible or VI-friendly in terms of quickly adjusting UI settings, such as kiosks, multimedia guides, and wearable displays.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools.**

Keywords

Assistive Technology; Adaptive UI; Visually Impaired Assistance; Context-Aware Computing; Conversational Interactions; Large Language Models; Multi-agent Systems; Ambient Computing



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UbiComp Companion '25, Espoo, Finland
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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754391>

ACM Reference Format:

Kaiyuan Tang, Mark Quinlan, and Youngjun Cho. 2025. Exploring LLM Agents as Seamless and Context-aware UI Adjusters for the Visually Impaired. In *Companion of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754391>

1 Introduction

Traditional electronic devices often neglect the personalised needs of visually impaired (VI) users as many of them tend to make their user interfaces (UI) one-size-fits-all, resulting in experiences that are not inclusive and accessible [21]. With the concept of Adaptive UI [6], many studies have been conducted to develop solutions to provide adjustable interfaces for VI users [2, 13, 21]. However, existing approaches largely rely on manual, cumbersome adjustments to UI settings or the use of external assistive tools such as magnifiers and screen readers, which are still facing context-awareness and VI-friendliness issues. This is particularly an accessibility challenge in scenarios involving public information terminals¹ [18].

With the development of Large Language Model (LLM)-driven Conversational Agents (CAs) [5, 20], this paper introduces a novel adaptive UI prototype, in which LLM CAs provide real-time UI adjustments to accommodate the sensory differences of VI users. The roles of CAs are bidirectional. First, they can digest conversational input and device and ambient context to understand the user's conditions and other environmental factors, infer the user's

¹An information terminal is a self-service, interactive computing device typically deployed in public spaces that provides users with access to digital information, services, or transactions through a user-friendly interface, commonly manifesting as kiosks, digital signage systems, or standalone computer stations designed to deliver specific informational content or facilitate particular tasks without requiring human assistance [17].

needs, and adjust the UI accordingly. Second, they can transform information from the UI into accessible voice output. This approach aims to better support different VI users than current solutions by transforming the interface from static to dynamic, and create seamless, context-aware, and VI-friendly user experience through natural conversational interactions (Figure 1).

The remainder of the paper is structured as follows. Section 2 reviews the background and related work. Section 3 details the system design, prototyping, and our initial tests². Finally, Section 4 concludes the paper and discusses implications and future work.

2 Background and Related Work

2.1 Challenges that VI Users Pose to UI Design

More than 285 million people are blind and partially sighted [4], there is a large group of VI users that often cannot fully benefit from using electronic devices [22]. the diversity of visual impairments leads to a wide range of special challenges [3, 22]. For example, some VI users might require larger text and bold font, some might need high-contrast colour schemes with dark background, while others might rely on assistive software to convert text to speech. Accommodating the needs is the key to enabling them to access digital information and services. As the number of digital information and services continues to grow, developing better accessibility solutions for VI users has become crucial to promoting inclusive and equitable use of digital devices [22].

Researchers and developers have explored the Adaptive UI paradigm, which is defined as a UI that can adapt its layout, style, or content in order to better fit the needs of the user or context [6, 7]. With this paradigm, some websites integrate accessibility overlays to allow VI users to manually adjust display preferences through the settings menu to change layout, font size, page colour, etc. Besides, built-in system magnification or assistive software (e.g. Microsoft's Immersive Reader) also allow VI users to zoom in, adjust line spacing, adopt colour schemes, or gain speech and non-speech auditory feedback from the UI [28].

However, each existing solution can only offer fragmented, limited accessibility options and requires complex setting setup that are not VI-friendly and depend on the user's technical skills. In shared terminals that prohibit the installation of personalised assistive software or preserve user profiles, VI users are more likely to be stuck with the vendor's fixed UI [19]. Even when adaptive features do exist, they often must be enabled by sighted staff and tend to be context-insensitive, for example, problems in device usage for VI users caused by situational factors such as changes in ambient light still lack a seamless way to adapt [27, 31]. Furthermore, industry practice has largely relied on the combination of separate adaptive features rather than developing a unified model [13], which leads us directly to our research gap: How do we combine known VI assistive features into a coherent, intuitive, and context-aware solution that covers dynamic usage scenarios?

2.2 LLMs' Roles in VI-friendly Adaptive UI

The recent development of LLMs presents potential opportunities to improve the experiences of VI users with accessibility features. LLMs show their ability in dialogue-based reasoning and problem-solving [1, 5, 35], as conversational interaction is considered one of the most user-friendly forms of human-computer interaction, LLM-powered CAs have already been studied as useful assistive tools to help VI users and other vulnerable groups [25, 30, 32, 34]. However, current research has not fully explored the potential of LLM-powered CAs as proactive action takers in UI adjustment. The latest LLMs can call tools based on user input and even generate interface elements [24, 29], making their functionality to VI users not limited to providing information, but also allows them to act as helpful UI adjusters. By processing user input, current device state, and ambient data from sensors [26], LLMs can transform unfriendly UI adjustment processes into intuitive, seamless, and conversational forms [11, 16]. For instance, when entering the UI, a user with moderate visual impairments can tell the CA that they want larger fonts and higher contrast colour scheme, the CA will proactively make changes suitable for the current device and ambient context without the need for manual adjustments; a user with serious vision loss can also speak to the CA to click buttons or inquire about the content on the UI. During use, the users can also engage in real-time, continuous chat with the CA for further adapting the interface.

The paper proposes that LLM-powered CAs have great potential to serve as powerful intermediaries between VI users and electronic devices to achieve Adaptive UI, and it can be a potentially good solution to unify various accessibility features and data sources of adaptation into one interaction framework.

3 Prototyping

3.1 System Design

To explore the proposed assistive features, we created a web-based prototype integrating LLM-powered CAs. The prototype includes a classic UI layout, a colour scheme, some demonstrative text content, and buttons that lead to different subpages. We set up an area on the side to enable user interaction with the LLM agent. Once the LLM agent is activated, users can speak with the agent, ask it questions or have it perform actions.

For the LLM agent, we adopted a modular design to enable the model layer to switch to different LLMs. We employ prompt engineering and system-level instructions to confine the LLM instances' functions to helping VI users know about the UI content or improve UI accessibility. We enabled the agent to access a context object, which includes the current state of the device and UI (e.g., screen size, UI location and theme), as well as ambient sensing data (e.g. light). We instructed it to utilise the context and conversation logs to fully understand the VI user's conditions and translate them into specific, addressable needs. Lastly, we designed a set of UI adaptation toolkit including text, layout, and colour scheme adjustments and tools that allow the agent to access the content the user is currently focusing on. We enable the agent to search for options in the preset toolkit that match the needs and confirm the actions with the user both before and after executing the tools.

In development, we noticed that the variety of user queries can easily lead to confusion in the agent's behaviour. To resolve this, we

²The complete code to the prototype described within this paper can be found at the following location: <https://github.com/PhysiologicAILab/llm-driven-adaptive-ui-for-the-visually-impaired>.

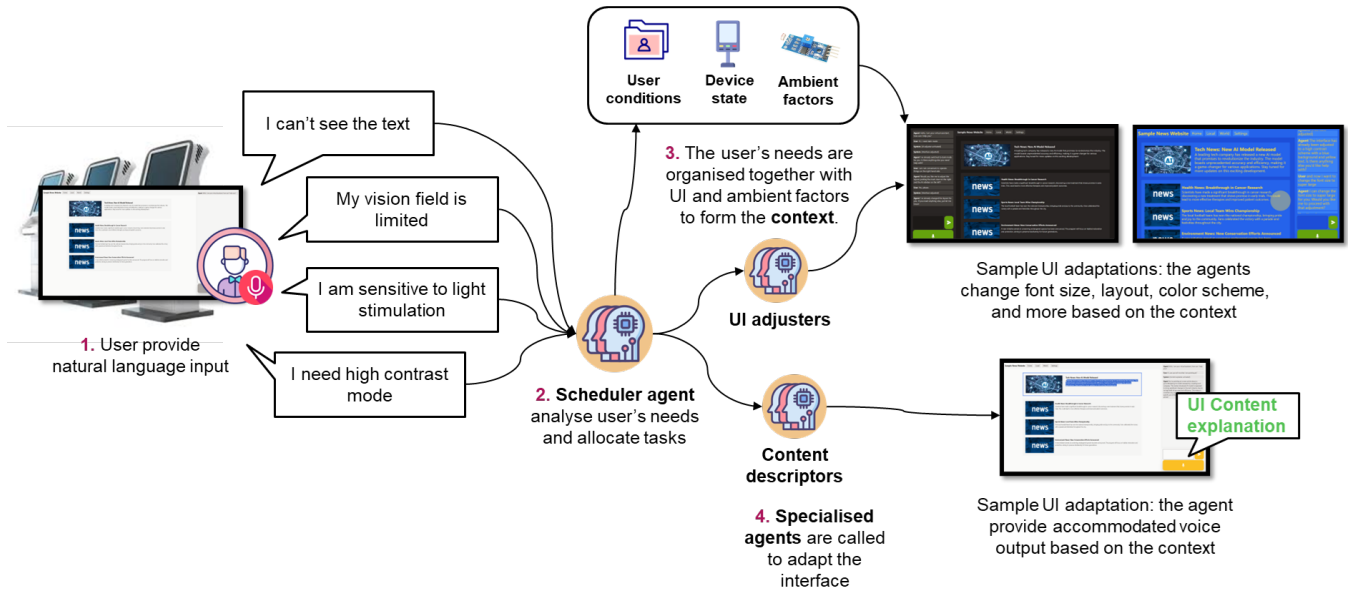


Figure 2: The system workflow and sample adaptations: The process begins with user input indicating needs such as difficulty of seeing text. A scheduler agent analyses these needs, allocates tasks, and organise user, device, and ambient contexts. Specialised agents, such as UI adjusters and content explainers, digest the context and adapt the interface accordingly. Examples of adaptations include changes in font size, colour scheme, layout, cursor visibility, and accommodated voice output.

adopted a multi-agent architecture inspired by OpenAI Swarm [14]: a single, specialised agent is trained to use a specific set of tools (e.g. colour adjustment) and prompted to only responsible for the specific category of user inquiries. A meta-agent (otherwise known as a scheduler agent) initially chat with the user, break down the user's queries into task scheduling plans, and divides out the tasks to corresponding specialised agents.

The workflow starts from a user entering the system, the scheduler agent will be available on the sidebar. If the user wants to adjust the UI, they simply need to have a voice conversation with the agent and express their needs. The agent will then infer and allocate the corresponding specialised agents and tools to help the user adjust the interface to a comfortable state. This dialogue process is continuous, all conversations and tool-calling logs are also used as context for subsequent interactions (Figure 2).

3.2 Initial Tests

To validate our design, we tested the agent with sample user profiles and user messages. We connected different LLMs such as Llama-2, Mistral, Claude-3.5, and GPT-4, to observe their responses and system behaviours. As a benchmark, the adaptations on Figure 1, 2, 3, and 4 are results using the GPT-4o model.

With the help of LLM CAs, VI users can easily transform text on the interface into conversational audio output and make UI adjustments by expressing their needs in natural languages. For VI users navigating or adjusting an unfamiliar UI, this approach demonstrates substantial potential to smooth the challenges they may face. Table 1 lists use cases with sample user inputs and the corresponding accommodating behaviours of the CAs in our initial tests.

user profile / message	CA behaviours
I can't see the text.	Provide emotional support; Increase the font size; Apply voice output.
I have amblyopia and am sensitive to light stimulation.	Recommend reducing brightness according to the current ambient light; Change the interface to dark mode; Apply voice output.
I have low visual acuity.	Provide emotional support; Increase the font size; suggest changing the interface to High-contrast mode according to the current ambient brightness.
I cannot see my mouse.	Enhance cursor visibility, suggest changing the interface to high contrast mode.
My right field of vision is limited.	Change the UI layout by moving the sidebar to the left side.
Can you tell me about the content on the page.	(With voice output turned on) naturally explain the information on the page to the user.
I want to go to [a subpage].	Navigate the user to the corresponding subpage.

Table 1: User input samples and corresponding agent behaviours: in addition to adjusting the interface, we also explored scenarios where the agent acts as a content narrator and assists users in performing tasks (e.g. navigating to a subpage) based on the current device state.



Figure 3: Illustration of a sample UI change case and the corresponding conversation transcription: the VI user tell the CA their visual condition, the agent then provide suggestions based on the user’s potential needs, the current device state and ambient context, and the adaptation tools available. The agent is capable of providing friendly and accurate suggestions and seamlessly adjust the UI to a new state. Further adjustments can be triggered via continuing the chat.

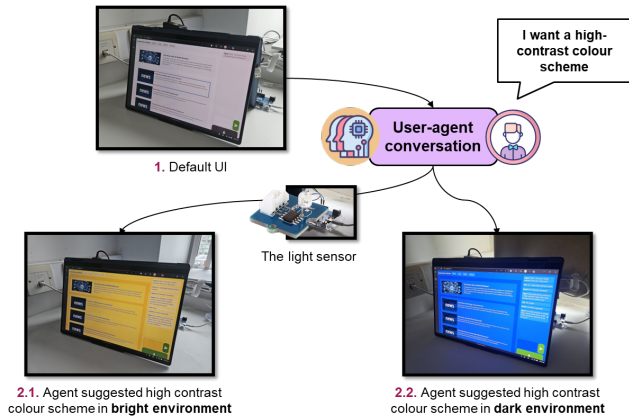


Figure 4: Adaptations on the sample information terminal with ambient factors: In making interface adjustments, the agent incorporates input from sensors to ensure that the adjustments are in harmony with the ambient lighting conditions and other factors.

The system incorporates environmental context such as lighting and device form factor to suggest appropriate adaptations proactively. We explored the integration of environmental context by using an Arduino light sensor as a source of input for ambient brightness. Based on this input, we instructed the agent to modified some adjustment behaviours. Figure 4 shows the different adjustment strategies chosen by the CA in response to user requests for a high-contrast interface in bright and dark environments, so that the adjusted interface aligns with ambient light. This paradigm aligns with ambient computing principles, where accessibility features emerge naturally from an intelligent environment response rather than explicit user configuration [12].

From the perspective of our lead author, who is a VI user, we observed that the integrated LLM CAs greatly enhanced the VI user’s interaction experiences. The author noted significantly lower use of external assistive tools such as magnifiers and screen readers throughout the process. While the conversation-based interface adjustment alleviated anxiety about being unable to read certain parts of the screen, with the agent providing voice output, and

making others visible to the point of ease of use. We therefore believe that in addition to physical accommodations, the system could also foster psychological inclusion by reducing helplessness and stress, possibly with personal sensing (e.g., [9, 23]). LLM CAs can be further fine-tuned and prompted to achieve a kind of ‘access intimacy,’ which means they can recognise the right adjustment and provide friendly and suitable suggestions and emotional support for VI users (Figure 3) [33].

4 Discussion and Conclusion

This paper explored the development of an LLM-powered prototype to create intuitive and context-aware Adaptive UI for VI users. Our approach enables real-time interface adaptations through natural language interaction, addressing the limitations of current manual configuration processes.

The prototype successfully implemented bidirectional conversational interaction where users can both request UI adaptations and receive audio descriptions of visual content. The system can make seamless and context-aware UI adjustments by digesting the VI user’s needs and device and ambient context. The multi-agent design proved effective, with specialised agents handling distinct adaptation tasks coordinated by a scheduler agent. Observation of usage with a VI-user demonstrated substantially lower use of magnifiers and screen readers during prototype interaction, noting reduced interface-related anxiety and increased autonomy, and establishes the technical feasibility and promise of LLM-driven adaptive interfaces. The approach shows particular promise for shared information terminals, kiosks, and wearable and portable displays where traditional accessibility configurations are unavailable and the diverse usage environments can easily create obstacles for individuals with low vision.

Further development will expand the prototype with more sophisticated adaptations, add more ambient sensor input (e.g., eye tracking [8], webcam or smartphone camera-based physiological sensing [10, 15]), integrate with existing UI platforms, and conduct comprehensive participatory studies with diverse VI users. We aim to evaluate quantitative metrics such as task completion time and adaptation accuracy alongside qualitative feedback on user satisfaction and accessibility improvements.

Furthermore, we plan to explore fine-tuned open-source models that can be deployed locally [36], addressing privacy concerns inherent in cloud-based LLM systems. This approach would be particularly valuable for public terminals and shared devices where users' accessibility needs and personal information should remain private and not transmitted to external servers, and in situations where network connectivity is poor.

As LLM capabilities advance, we anticipate more nuanced and context-aware adaptations that could transform how assistive technologies integrate with mainstream computing environments. This work contributes to the foundation for truly inclusive digital experiences where accessibility emerges from intelligent system adaptation rather than separate assistive tools.

Acknowledgments

This work was supported by the Research Methods and Making Skills module (COMP0145), part of the Disability, Design and Innovation MSc programme at University College London.

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